

Introduction to Machine Learning - Lecture 1

Machine learning (ML) is the study of algorithms that improve their performance on a task through experience, usually in the form of data. Instead of writing explicit rules, we let a model discover patterns in examples.

Three main paradigms are distinguished. In supervised learning the training data contains input-output pairs and the goal is to predict the output for unseen inputs. In unsupervised learning only inputs are available and the goal is to discover structure, for example clusters. In reinforcement learning an agent learns by receiving rewards for its actions.

Typical supervised tasks are classification (predicting a category, e.g. spam or not spam) and regression (predicting a number, e.g. a house price). The k-nearest neighbours algorithm, decision trees, logistic regression and neural networks are common supervised methods.

Training, validation and testing

A data set is normally split into three parts. The training set is used to fit the model parameters. The validation set is used to choose hyperparameters such as the learning rate or the depth of a tree. The test set is used exactly once, at the end, to estimate how the model will perform on new data.

Overfitting happens when a model learns the noise in the training data instead of the underlying pattern. Symptoms are a very low training error combined with a much higher validation error. Common remedies are collecting more data, regularisation, early stopping and simpler models.

Underfitting is the opposite problem: the model is too simple to capture the pattern, so both training and validation errors stay high. The balance between the two is often described as the bias-variance trade-off.

Cross-validation repeats the train/validation split several times (for example 5-fold) and averages the results, which gives a more reliable estimate when the data set is small.

Evaluation metrics

For classification the confusion matrix counts true positives (TP), false positives (FP), true negatives (TN) and false negatives (FN). Accuracy is $(TP + TN) / \text{total}$. Precision is $TP / (TP + FP)$ and measures how many predicted positives are correct. Recall is $TP / (TP + FN)$ and measures how many actual positives were found.

The F1 score is the harmonic mean of precision and recall: $F1 = 2 * \text{precision} * \text{recall} / (\text{precision} + \text{recall})$. It is useful when the classes are imbalanced, because accuracy alone can be misleading: a model that always predicts 'healthy' is 99% accurate on a data set with 1% sick patients but is useless.

For regression the mean squared error (MSE) averages the squared differences between predictions and true values, and the mean absolute error (MAE) averages the absolute differences. The coefficient of determination R^2 tells how much of the variance is explained by the model; 1.0 is perfect.

Course note: the final exam is on 12 December and covers lectures 1 to 8. The verification code for this handout is ML-2026-ALPHA.